

TEACHING PHILOSOPHY

BUILDING BRIDGES TO LEARNING

I believe effective teaching begins with a simple premise: students learn best when they are given opportunities to think, try, question, and try again. My role as an instructor is not simply to deliver information, but to create conditions for students to develop understanding, confidence, and independence. I think of teaching as building a bridge between where students begin and where they can go. Each pillar of that bridge is the foundation of my teaching philosophy and includes 3 main pillars: (1) failing forward through active learning, (2) building a classroom community, and (3) technology that supports thinking. My goal as an educator is not to carry students to the other side, but to help them develop the knowledge and skills needed to cross it independently.

Pillar I — Failing forward through active learning

I believe learning should be active. Evidence from undergraduate STEM education shows that active learning improves examination performance and reduces failure compared with traditional lecture-based teaching (Freeman et al., 2014). This is especially important in biology, where students must move beyond recalling terminology to interpreting evidence and connecting genetics, evolution, behaviour, and physiology. In my teaching, I use questions, short problems, discussion, and applied examples to make students participants in the learning process. When teaching genetics, for example, I ask students to work through inheritance problems and pedigrees and explain their reasoning rather than simply reproduce definitions. In quantitative contexts, I ask students to interpret figures and statistical outputs and connect them back to a biological question; integrating quantitative reasoning into biology curricula has been shown to improve students' ability to interpret statistical analyses (Goldstein & Flynn, 2011).

I also believe that students need room to fail productively. Research on productive failure suggests that attempting unfamiliar problems before receiving formal instruction can lead to stronger conceptual understanding and transfer to new problems (Kapur, 2014). For this reason, I value low-stakes and ungraded activities. Students should be able to make an error in a genetics problem, struggle with an unfamiliar analysis, or offer an incomplete explanation without feeling that experimentation has permanently affected their grade. These activities allow me to identify misconceptions early while helping students reframe difficulty as part of learning: from "I can't do this" to "I don't understand this yet, but I know where to begin."

Pillar II — A classroom community built on trust, feedback, and inclusion

The second pillar of my bridge is the classroom community. I strive to create a learning environment grounded in kindness, respect, transparency, and curiosity, where questions and uncertainty are treated as normal parts of academic work. I establish clear expectations, invite students to contribute to a shared classroom culture, and aim to make participation feel intellectually safe without lowering academic expectations. This approach aligns with the Community of Inquiry framework, which emphasizes the interaction of teaching, cognitive, and social presence in meaningful learning (Garrison et al., 2000).

Assessment is part of this environment. I see assessment not only as a measure of what students know, but as a tool for moving learning forward. I use formative questions, daily check-ins, practice activities, and "ticket out the door" reflections that ask students to identify the clearest and muddiest points from a lesson. This gives students a chance to monitor their own understanding and gives me immediate evidence about where teaching needs to change. A large meta-analysis found that feedback has a meaningful overall effect on learning, while also showing that its value depends on the information it

provides (Wisniewski et al., 2020). I therefore use student feedback as instructional data: I listen, reflect, and adjust.

Pillar III — Technology that supports thinking

As a researcher who works with R, Python, statistical modelling, and genetic data, I want students to develop computational and statistical literacy alongside biological reasoning. I emphasize not only how to use a tool, but why an analysis is appropriate, what its assumptions mean, how to recognize limitations, and how to explain a result in biological terms. I also see generative AI as a technology that can be incorporated into active learning when its use is transparent and critically evaluated. Students might examine an AI-generated explanation or piece of code, identify errors or missing context, and verify the result against course materials and scientific evidence. Current scholarship suggests that generative AI can support learning, but that pedagogical scaffolding and clear instructional guidance are essential (Liu & Zhong, 2025).

The other side of the bridge

Ultimately, I want students to leave my classroom with more than a collection of facts. I want them to analyze evidence, interpret data, ask thoughtful questions, communicate scientific ideas, and approach unfamiliar problems with confidence. I use Universal Design for Learning as a guiding principle for reducing unnecessary barriers through clear organization, accessible materials, and multiple opportunities to engage with and demonstrate learning; UDL has been identified as a useful framework for planning more accessible higher-education teaching (Cumming & Rose, 2022).

Good teaching, to me, is therefore both rigorous and humane. It asks students to do difficult intellectual work while giving them the structure and support needed to succeed. It also requires me to keep learning. I continue to develop my teaching through workshops, professional development, reflection, and feedback from students and colleagues. The bridge is never finished: I refine its pillars as I learn what helps students move from uncertainty to understanding, and from dependence on an instructor to confidence in their own ability to learn.

References

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